08 LECTURE HEALTH CARE

MT 1, Second Semester LECTURE 08 - Mr. Andrae D. Reyes, RMT DTA

DISASTER CONTROL MANAGEMENT

TOPIC OVERVIEW

LEARNING OBJECTIVES

I. Learn terms that define and constitute a disaster
II. Identify the different types of hazards and disasters
III. Understand how vulnerable the Philippine archipelago is by knowing disasters in the Philippine setting
IV. Discuss how disasters are managed through implemented plans and measures

I. TOPIC 1

- Although a **disaster** is considered an **emergency**, an **emergency** alone does not necessarily mean that it has to be a **disaster**.

DISASTER

- Depending on the characteristics of the disaster itself, when we talk about a disaster, this is an **emergency** that **goes beyond the ability** of the community to **respond** as well as **recover** from the incident using their own resources.
  - **Frequently requires assistance** from outside the immediate community, both to **manage resulting issues** and to **recover completely**.
- Any event that causes a level of destruction, death, or injury that affects the abilities of the community to respond to the incident using available resources
- It affects the ability of the community to respond to the incidents using available resources.
- A disaster is usually a **cause** and **effect**
  - Causing mass casualties, multiple casualties
    - **Mass casualty**: 100 or more
    - **Multiple casualty**: more than 2 but less than 100
  - **Affecting**:
    - Direct victims (including: displaced persons, refugees),
      - **Displaced persons:** are does individual who have to **evacuate their home,** school, or even business as a result of a disaster
      - **Refuges**: These are a group of people who **fled their home** or country as a result of famine, drought, natural disaster, war, or even a form of civil unrest.
  - **Indirect victims** (such as the relatives of the direct victims) can cause **psychological** effects)
- **WORLD HEALTH ORGANIZATION (WHO)**
  - Disaster is "Any occurrence that causes damage, ecological disruption, loss of human life, deterioration of health and health services on a scale sufficient to warrant an extraordinary response from outside the affected community or area."
- A disaster resulted from the combination of **hazard**, **vulnerability**, and **insufficient**

capacity or measures to reduce the potential chances of risk.

- **Formula**:

$$\text{DISASTER} = \frac{\textit{HAZARD} + \textit{VULNERABILIT}}{\textit{CAPACITY TO COPE}}$$

BREAKDOWN

- HAZARD + VULNERABILITY **= RISK** .
- **CAPACITY TO COPE**
    - **Insufficient capacit**y: If an individual has a ↓ **lower capacity to cope,** therefore they are unable to manage the **disaster.**

EMERGENCY

- An emergency is defined as an individual who can manage their own situation **using their own resources.**
- An emergency is any event endangering the life or health of a significant number of people and demanding immediate action. Situations may arise due to:
    - natural
    - man-made
    - technological
    - societal hazards

HAZARD

- Any phenomenon that **has the potential to cause disruption or damage** to people, their properties, their services, and their environment.
- A hazard can be man-made, but most hazards are perceived as natural
- **Disaster** happens when a hazard impacts the vulnerable population and causes damage, casualties, and disruption.
- Without a hazard affecting the vulnerable population, these does not constitute a disaster

- **Differentiate hazards from disasters:**
    - A **hazard** is an **event**, and the **disaster** itself is the **consequence**.
    - A **hazard** is perceived as an event that **threatens life and property**, while a **disaster** is the **realization of these hazards.**
    - Hazards may be **inevitable**.
    - Disaster can be **prevented**.

Disaster may result from natural, biological, technological, or societal hazards.

- **Natural Hazards**
    - Physical force, such as typhoon, flood, landslide, earthquake, and volcanic activity.
- **Biological Hazards**
    - a process or phenomenon of organic origin (infectious agents) or conveyed by biological vectors, including exposure to pathogenic microorganisms, toxins, and bioactive substances.
- **Technological Hazards**
    - arise from technological or industrial conditions, including accidents, dangerous procedures, and infrastructure failures.
- **Societal Hazards**
    - result from the interaction of varying political, social, or economic factors, which may have a negative impact on the community
        - Wars, conflicts, nuclear wars
- **Complex Hazards** (multiple hazards in one)
    - Natech Hazards
        - Natural-technological disaster
        - This type of disaster constitutes more than 2 and is affected by or caused by 1 natural hazard and 1 technological hazard.
        - **Natural Hazard Triggered**
            - Earthquake (Natural) It causes the faulty wiring to spark and lead to fire (technological)

A. TYPES OF HAZARDS AND DISASTERS

CLASSIFICATION OF NATURAL AND TECHNOLOGICAL DISASTERS

Disaster Group	Disaster Sub-group	Disaster Main Type	Disaster Sub - type
NATURAL	Geophysical	Earthquake	Ground movement, Tsunami
		Mass movement (dry)	Rock Fall, Landslide
		Volcanic Activity	Ash Fall, Lahar, Pyroclastic flow, Lava flow
	Meteorological	Storm	Extra Tropical Storm, Tropical Storm, Convective storm
		Extreme Temperature	Cold wave, Heat wave, Severe winter condition
		Fog	
	Hydrological	Flood	Costal Flood, Riverine Flood, Flash Flood, Ice Jam Flood

		Landslide	Avalanche (snow, debris, mudflow, rockfall)
		Wave action	Rogue Wave, Seiche
	Climatological	Drought	
		Glacial Lake Outburst	
		Wildfire	Forest Fire, Land Fire: Bush, Pasture
	Biological	Epidemic	Virus, Bacterial, Parasitic, Fungal, Prion Disease
		Insect Infestation	Grasshopper, Locust
		Animal accident	
	Extraterrestrial	Impact	Airburst
		Space Weather	Energetic particles, Geomagnetic storm, shockwave
Technological	Industrial accident		Chemical spill, Collapse, Explosion, Fire, Oil spill, Poisoning, Radiation, Gas leak
	Transport accident		Air, Road, Rail, Water
	Miscellaneous accident		Collapse, Explosion Fire
	Sociological		Stampede, armed conflicts, terrorist activities, Riots

Table 1. *List of Natural and Technological Disasters*

Table 1 Discussion

- **Geophysical**
  - **Geo =** Land
  - **Physical** = Movement
- **Meteorological**
  - Involves the atmosphere
- **Hydrological**
  - Involves water
- **Climatological**
  - It does touch the atmosphere part, but rather more on changes in temperature of a certain geographic location that promotes/leads to a disaster.
- **Extraterrestrial**
  - This involves disasters that do happen outside the planet (space-bound)
  - **Ex**: Impact - **Airburst**

Sudden impact	Earthquakes, Tropical storms, Tsunamis, Volcanic eruptions, etc.
Slow onset	Drought, Famine, Pest infestation, Deforestation, etc.
Epidemic DIsease	Water-borne, Food-borne, Vector-borne, etc.

Table 2. *Types of Duration and Their Examples*

Table 2 Discussion

- **Sudden impact**
  - The immediate effect cannot be predicted
- **slow onset**
  - Delayed effect, you cannot really tell unless you've noticed some differences.
- **Epidemic diseases**
  - This can be a sudden or slow onset
  - There are epidemics that are gradually rising or slower.
  - Caused by a biological agent

II. VULNERABILITY, RISK, AND CAPACITY

A. VULNERABILITY

- The extent to which a community or structure, services, or geographic area wherein they are more likely to be damaged or disrupted by the impact of a particular hazard on account of their nature, destruction, as well as proximity to hazardous terrains or disaster-prone areas.
- conditions or set of conditions that reduce the people's ability to prepare for, withstand, or respond to a hazard
- The propensity of things to be damaged by a hazard
- You can't call a disaster if it's just a hazard alone; it needs to have a vulnerable population to be called a disaster.
- **CATEGORIZED INTO:**
  - **Physical** - who or what may be destroyed (Ex, unstable infrastructures, disabled people)
  - **Social** - inability of the affected population to endure the unfavorable impacts of a hazard
    - Includes social aspects of literacy, good governance, social justice, conventional values, customs, and ideological beliefs
      - **Literacy** - illiterate people have a hard time comprehending
      - **Media illiteracy** - prone to believe

AI spliced videos or fake news
- **Governance** - the better the governance, the lower the social vulnerability
- Ideological beliefs - an example is how you can transfuse blood to a patient who is a Jehovah's Witness; basically, respect the patient's beliefs

○ **Economical** - businesses and assets

○ **Environmental** - includes resource degradation, aftermaths of famine or droughts (which lessen agriculture), and death of livestocks
  - The more our resources lessen, the more vulnerable the population will be

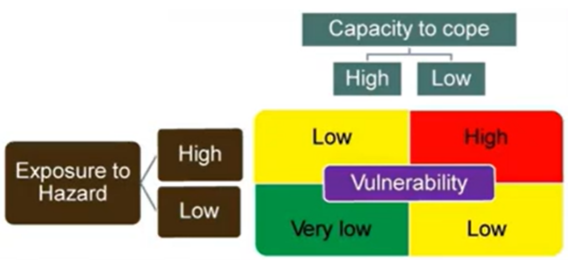

Figure 1

Figure 1 Discussion

- Low Capacity to cope + High Exposure to hazard
  - You will not have enough resources to manage the effects of the hazard
- High Exposure to Hazard + High Capacity to cope
  - Since you have the means to manage your situation, it makes your vulnerability is low
- Low Exposure to Hazard + Low Capacity to cope
  - Your vulnerability is low
- Low Exposure to hazard + high Capacity to Cope
  - Your vulnerability becomes very low since you have the means or resources to aid yourself.

B. RISK

- Is a situation involving exposure to danger or the possibility that something unpleasant or unwelcome will happen
- A probability that a community's structure or geographic area is to be damaged or disrupted by the impact of a particular hazard, on account of its nature, construction, and proximity to a hazardous area

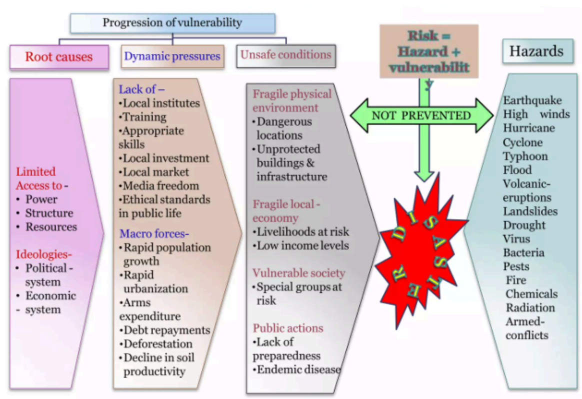

Table 3. Progression of Vulnerability

Table 3 Explanation

- Root causes are the factors involved.
- Dynamic pressures are created by the root causes
- Unsafe conditions or events that may take place
- *Risk = hazard + vulnerability*
- In the context of disaster management, risk is viewed as the product of potentially damaging events and vulnerable conditions of a society or elements exposed.

Why are disasters increasing?

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1. **Increase in population** - increases the amount of people vulnerable.
2. **Climate Change** - the more increase in temperate, the more are the chances that the impacts of disasters worsen.
3. **Increased vulnerability due to :**
   a. **Demographic changes** - (older/younger are more vulnerable)
   b. **Increased concentration of assets** - more assets = higher possibility of destruction
   c. **Environmental degradation** - the more our environment worsens, the higher the impact of the disaster will be
4. **Poverty** - if you don't have or have less capacity to cope the impact of the disaster will be increased on those people
5. **Rapid urbanization and unplanned development**
   - Those situations or areas that were improperly planned may lead to a disaster awaiting
   - **Rapid urbanization** - concentrated population means there will be more affected by the disaster.

C. CAPACITY

- Resources, means, and strengths which exist in a household or a community which enables them to cope with, withstand, prepare form and prevent, mitigate, or quickly recover from a disaster
- Positive conditions or abilities which increase a community's ability to deal with hazards
- **TWO TYPES**:
    - **Physical Capacity**
        - **Man power**
        - **Strong Infrastructure**
    - **Socio-economic Capacity**
        - **Connections and privileges** - the more connecting or privileges you have, the more you will be able to recover quickly from a disaster.

III. DISASTER IN THE PHILIPPINE SETTING

- The Philippines is considered as one of the most vulnerable countries in the world considering its numerous hazards present in the country.
- Lies in the Pacific Ring of Fire; 200 volcanoes, 22 are considered active and destructive (5 are closely monitored).
- The country lies along the **Typhoon Belt** - average of 24 typhoons/year.

Characteristics

- **Frequency**
    - How often does a disaster occurs.
- **Predictability**
    - Ability to tell when and if a disaster event will occur.
- **Preventability**
    - Actions can be taken in order to avoid a disaster.
- **Imminence**
    - Immediate, speed of onset of an impending disaster, and related to the extent of forewarning possible and the anticipated duration of the incident.
    - Ex. How sudden, gradual, and hazardous the disaster is.
- **Scope and Number of Casualties**
    - The range of the effects of the disaster.
    - The bigger the scope, the wider the range of the disaster occuring. The number of casualties will also increase.
- **Intensity**
    - Level of destruction and devastation of the disaster.
    - How intense the disaster is.

EARTHQUAKE PRONE AREA		
Region 1 - Ilocos Norte - Ilocos Sur - Pangasinan - La Union	**Region 6** - Antique - West Panay - Negros Occidental - Iloilo	**Region 12** - South Cotabato
Region 2 - Batanes - Cagayan	**Region 7** - Cebu - Bohol - Negros Oriental	**ARMM** - Maguindanao - Sultan Kudarat
Region 3 - Bataan - Bulacan - Zambales - Nueva Ecija	**Region 8** - Eastern Samar - Leyte - Northern Samar	**CARAGA** - Surigao Del Sur - Surigao Del Norte
Region 4 - Mindoro	**Region 9** - Zamboanga	**CAR** - Benguet
- Marinduque	Provinces	- Baguio City
Region 5 - Albay - Catanduanes	**Region 10** - Bukidnon	

IV. DISASTER MANAGEMENT

- Is a term used for a continuous and integrated process of **planning, organizing, coordinating, and implementing measures** which are necessary or expedient for:
    - Prevention of danger or threat of any disaster
    - Reduction of the risk of any disaster or its severity, or its consequences
    - Capacity-building
    - Preparedness to deal with any disaster
    - Prompt response to any threatening disaster situation or disaster
    - Assessing the severity or magnitude of effects of any disaster
    - Evacuation, rescue, and relief
    - Rehabilitation and reconstruction

- Disaster Management is the "umbrella term". This constitutes a broad range of activities.
- It encompasses a lot, it doesn't just encompass disaster risk management, it also involves **mitigation, preparation, recovery, response and even rehabilitation and reconstruction**

A. DISASTER RISK MANAGEMENT

A broad range of activities designed to:

- Prevent loss of lives, minimize human suffering, inform the public and authorities of risk, minimize property damage and economic loss, speed up the recovery process
    - Focuses on activities

- Part of disaster management
- Actions that can be done to manage the risk of a disaster

B. 16 AIMS OF DISASTER MANAGEMENT

- To reduce the impact of disasters and quantum of loss.
- To create an environment where individuals and communities work together in groups and are able to achieve selected aims effectively and efficiently.
- To develop important strategies to reduce and control the occurrence of disasters.
- To train individuals and communities to remain prepared for sudden disasters.
- To trigger the affected regions and the community's emergency resources for quick response.
- To coordinate and communicate for proper management of resources, namely, man, material and economic resources available for the purpose of disaster response and recovery.
- To foster team spirit, where persons rise above themselves to help the victims of disasters in whatever way they can.
- To organize recovery and rescue mechanisms.
- To generate resources necessary for rescue, recovery and post-recovery work.
- To elicit action for management of disaster in a time-bound manner.
- To facilitate the non-governmental and governmental machinery to work in tandem for disaster management.
- To commit resources for disaster mitigation, preparedness, rescue and recovery.
- To draw the attention of national and international agencies for disaster relief.
- To formulate policies for curbing the menace called disaster before its onset.
- To develop a systematic approach to the management of disasters.
- To foster local resilience to disasters by adopting a consensus-building approach in consultation with the local community.

C. DISASTER MANAGEMENT CYCLE

- **Preparation, Response, Recovery** and **Mitigation** are all aspects necessary for **disaster management**.
    - Management of disasters is a prerequisite, which incorporates the steps of all these four.
    - The **aim** of the disaster management cycle is to minimize the risk of disasters and to handle them when they do occur in an effective manner to limit and reduce the effect of loss.

1. **Response**
    - These are the measures required in search of survivors as well as to meet the basic needs for shelter, water, food, and akso health care.
    - This includes actions taken to save lives, prevent damage to property and to preserve the environment during emergencies or disasters.
    - This is where implementation of actions plans take place
    - Whatever is the action plan being prepared during preparation, it must be implemented during **disaster response**.
2. **Recovery**
    - Process undertaken by a disaster affected community to fully restore itself to pre-disaster level of functioning
    - Three terms that are common in disaster management:

- **<u>Relief</u>**
    - These are actions taken immediately following the impact of a disaster(e.g. relief goods)
    - **Focus on Basic needs** are given such as **food, water, clothing, shelter.**
- **<u>Rehabilitation</u>**
    - Actions that assist the community to restore to a sense of normality after a disaster.
    - More focused on the **people.**
- **<u>Reconstruction</u>**
    - These are actions taken in the aftermath of a disaster to assist victims to prepare their dwellings, reestablish essential services and revive key economic and social activities.
    - More focused on **structures.**

3. **Mitigation**
    - Measures taken prior to an impact of a disaster to minimize its effects.
    - This refers to the structural and non structural measures
    - This includes any activities that reduce either the chance of a hazard place or a hazard turning into a disaster.
    - Also includes **building codes, zoning and land use management, regulation and safety codes** and **preventive healthcare and public education.**
    - **Risk reduction** or **anticipatory measures** that seek to avoid future risks as a result of a disaster.

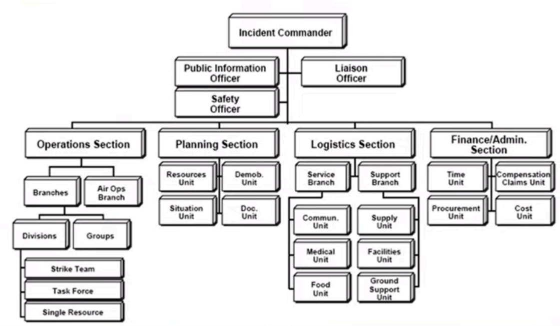
4. **Preparation**
    - Measures taken in anticipation of a disaster to ensure that appropriate and effective actions are taken in the aftermath.
    - **Plans** that are Made to saves lives or properties and help the response and rescue service operations
    - This phase covers **implementation, operations, early warnings, capacity buildings**
    - Also includes measures taken to avert a disaster from occurring(if possible).

V. PHASES OF DISASTER MANAGEMENT

A. DISASTER PREPAREDNESS

- Is an **ongoing, multi-sectoral activity**
    - **Multi-sectoral activity** - Involves not just one, but multiple levels of society:
        - ➢ Local (barangay, city)
        - ➢ Regional
        - ➢ National
        - ➢ International
- **Integral part of the national system** responsible for developing plans and programs for:
    1. **Prevention**
    2. **Mitigation**
    3. **Response**
    4. **Rehabilitation**
    5. **Reconstruction**
        - Basically the entire disaster cycle starts here
- It is **responsible for the coordination of sectors** to carry out:
    - Evaluation of the risk
    - Adopt standards and regulations
    - Organize communication and response mechanism
    - Ensure all resources are ready and easily mobilized
    - Develop public education programs
    - Coordinate information with news media
    - Disaster simulation exercises
        - Fire drills
        - Earthquake drills
- **Medical Preparedness & Mass Casualty Management**
    - Development and capacity building of medical team for Trauma & psycho-social care, mass casualty management and Triage
    - Determine the casualty handling capacity of all hospitals
    - Formulate appropriate treatment procedures
    - Involvement of private hospitals
    - Mark would be in care centers that can function as a medical unit
    - Identify structural integrity and approach routes

B. INCIDENT COMMAND SYSTEM (ICS)



- A **hierarchical chain of command** is used in disaster situations
- **Structure:**

1. **Command Staff**
    - In an incident command organization, which typically includes:
        a. **Incident Commander** (topmost) - overall leader
            - **responsible for all incident activities**, including the **development of strategies and tactics,** and the **ordering and release of resources**
            - has **overall authority and responsibility** for conducting incident operations and is **responsible for the management** of all incident operations at the incident site.
                - can be *president* (national view), *mayor* (city level), *barangay captain* (barangay level)
        b. **Public Information Officer** – responsible for **interfacing with the public and media** and/or with other agencies with incident-related information requirements
        c. **Safety Officer** – **monitors incident operations** and **advises the IC/UC on all matters** relating to *operational safety*, including the *health and safety of emergency responder personnel*
        d. **Liaison Officer** – is **Incident Command's point of contact** for DRRMC/other government agency representatives, NGOs, and the private sector, to **provide input on their agency's policies, resource availability, and other incident-related matters**

2. **General Staff**
    - a **group of incident management personnel** organized according to

function and reporting to the **Incident Commander**

- normally consists of the:

a. **Operations Section**

- responsible for all tactical activities focused on **reducing the immediate hazard**, **saving lives and property**, **establishing situational control**, and **restoring normal operations**
- OS is composed of:
  - ➔ **Branches** - may be **functional, geographic, or both**, depending on the circumstances of the incident
  - ➔ **Divisions and Groups** are established when the **number of resources exceeds the manageable span of control** of *Incident Command* and the *Operations Section Chief.*
    - ◆ **Divisions** - established to 10 divide an incident into physical or geographical areas of operation.
    - ◆ **Groups** - established to divide the incident into functional areas of operation
  - ➔ **Resources**
    - - **Single Resources:** Individual personnel or equipment and any associated operators
    - - **Task Forces:** Any combination of resources assembled in support of a specific mission or operational need
    - - **Strike Teams:** A set number of resources of the same kind and type that have an established minimum number of personnel

b. **Planning Section**

- collects, evaluates, and disseminates incident situation information and intelligence to the IC/UC and incident management personnel
- Within the Planning Section, the following primary Units fulfill functional requirements:
  - ➔ **Resources Unit:** records the status of resources committed to the incident
  - ➔ **Situation Unit:** responsible for the collection, organization, and analysis of incident status information, and for the analysis of the situation as it progresses
  - ➔ **Demobilization Unit:** ensures orderly, safe, and efficient demobilization of incident resources
  - ➔ **Documentation Unit:** collects, records, and safeguards all documents relevant to the incident
  - ➔ **Technical Specialists:** Personnel with special skills that can be used anywhere within the ICS organization

c. **Logistics Section**

d. **Finance/Administrative Section**

- An **Intelligence/Investigation Chief** may be established, if required, to meet incident management needs.

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C. DISASTER IMPACT

- **Consequences of Disaster**
  - **Health:**
    - ▪ **Physical** - entanglement, injuries, disabilities, coma, death
      - trapped in debris

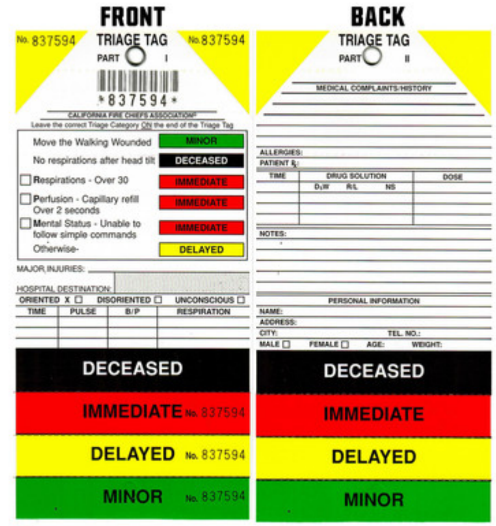
- **Psychological** - cognitive, behavioral, social

- **Types of Damages:**
  - **Structural damage** - variable extent
  - **Ecological** - changes in the ecosystem and damage to nature
  - **Economical** - financial losses

D. DISASTER RESPONSE

- **Immediate reaction to disaster** as the disaster is anticipated, or soon after it begins, in order to **assess the needs**, r**educe the suffering**, **limit the spread and consequences of the disaster**, **and open up a way to rehabilitation.**
- It is done by:
  - **Mass evacuation**
    - organized movement to safe areas
    - hectic evacuation = more casualties
  - **Search and rescue**
    - locate and save victims
  - **Emergency medical services**
    - must always be accessible in terms of response to an immediate disaster.
    - ambulances, field hospitals, first aid
  - **Securing food and water**
    - Basic needs like food, water, clothing, shelter
  - **Maintenance of Law & Order**
    - prevent looting, violence, and chaos
- **Disaster Impact & Response are involved in the:**
  - **Search, Rescue, and First Aid**
    - Those individuals involved in search and rescue operations are mostly trained in first aid.
  - **Field care**
    - Part of the job of a safety officer is to assess whether or not the scene is safe for the search and rescue to go to a certain location
  - **Triage**
    - Prioritization of the victim
  - **Tagging**
    - the act of labeling them
  - **Identification**
    - Assess who should be attended to first
- **Disaster Management: Triage**
  - Categorizes victims into four categories depending on the urgency of their needs for evacuation, as well as the treatment or care they need
  - Maximize survival when resources are limited.
  - In a disaster or mass casualty situation, different systems for triage have been developed.
  - One system is known as **START** (Simple Triage and Rapid Treatment)
    - It categorized the victims into 4 categories depending on the urgency and the care needed for the victim.
  - If necessary, **START** can be implemented by persons without a high level of training.

- **TRIAGE COLOR TAGS**



- **Red = Immediate**
  - Those who **cannot survive without immediate treatment, but who have the chance of survival**
  - Those who are injured and can be helped by immediate transportation
- **Yellow = Delayed**
  - **Stable for the moment**, and they are **not in immediate danger of death**
  - Individuals who are injured with less severe injuries whose transport can be delayed
- **Green = Minor**
  - **"Walking wounded"** who will need medical care at some point, after more critical injuries have been treated
  - Those with minor injuries not requiring urgent care
- White
  - Those with **minor injuries**
- **Black = Deceased/Beyond help**
  - **Deceased** and for those whose **injuries are so extensive** that they

will not be able to survive, given the care that is available
  - Not everyone can be saved
  - Priority = greatest number of survivors
  - Other systems, not just START, can be used in mass casualty situations
  - **Tags can change depending on the condition:**
    - **Green → Yellow → Red (remove previous tags as condition worsens)**
- **Medical and Public Health Response**
  - Pre-hospital emergency services
    - Linkage to the government. incident command system
    - External medical services & extrication workers
    - Search and Rescue Teams
  - Assessment of immediate health needs
  - Identification of medical & health resources
  - Temporary field treatment
    - Followed by a prompt and proper treatment to save lives

Should be looked into:
- Food safety and water supply
- Animal control
  - Carcasses can foul water
  - Zoonotic diseases
- Vector control
  - Mosquito and rodents communicable disease control: measles, diarrheal diseases, ARI, and malaria
- Breakdown in environmental safeguards
- Crowding of persons in camps
  - Malnutrition
  - Waste management
- Temporary latrines
  - Chemical toileting
  - Sewage disposal damage
- Management of hazardous agent exposure
- Also infectious agents if hospital or scientific laboratories damaged
- Mental health services
  - Specialized psychological triage and treatment (significant in terrorism)
- Information
- Behavioral contagion handling
- Risk communication

E. DISASTER RECOVERY

- Long-term process after a disaster
- Includes the "4 Rs": **Relief, Rehabilitation, Reconstruction,** and **Repatriation**
- **Repatriation**
  - After the emergency is over, displaced people return to their place of origin
- **Rehabilitation**
  - Restoration of Basic Social Functions
  - Providing Temporary Shelters
  - Stress Debriefing for Responders and Victims
  - Economic Rehabilitation
  - Psycho-social Rehabilitation
  - Scientific Damage Assessment
- **Elements of Recovery**
  1. Community Recovery
  2. Infrastructure Recovery
  3. Economic Recovery
  4. Environmental Recovery

F. DISASTER MITIGATION

- The **measures taken prior to the impact of the disaster to minimize its effects** (sometimes referred to as ***structural and non-structural measures***).
  - Any activity that reduces either the chance of a hazard taking place or a hazard turning into a disaster
- Includes:
  1. Building codes
  2. Zoning and land use management
  3. Regulation and safety codes
  4. Preventive health care

  Public education

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